Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Electrostatic Discharge (ESD) Control for Lab and Factory Engineers and Managers

TGS Ref No: TGS-2024051412

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 3.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
2.0	Legacy release	Legacy two-day outline.	Tertiary Infotech Academy Pte Ltd
3.0	19 August 2026	Rebuilt two-day plan aligned to the v3.0 slide map, eight activities and WA plus Case Study.	Tertiary Infotech Academy Pte Ltd

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Course Overview

Two-day WSQ programme. TSC: Electrostatic Discharge Control (ELE-OPR-4002-1.1). Assessment: Written Assessment (SAQ) — 4 open-ended questions, K1–K4, 1 hour, open book. Case Study (CS) — 2 open-ended tasks, A1–A3, 1 hour, open book.

Learning Outcomes

- LO1: Apply control measures to manage ESD generation based on ESD principles.
- LO2: Evaluate and recommend actions to improve ESD control effectiveness in accordance with ESD standards.

Day 1 Schedule

Time	Session / learning activity	Slide reference
09:00–09:30	Digital attendance, introductions and course briefing	Admin / assessment
09:30–10:30	Module 1: charge generation, discharge paths and models	15–105
10:30–10:40	Morning break	Admin / assessment
10:40–12:15	Failure modes, exposure points and business impact	15–105
12:15–13:00	Activities 1–2: hazard hunt and failure triage	15–105, A1: 99–101, A2: 102–104
13:00–13:45	Lunch	Admin / assessment
13:45–15:15	Module 2: EPA, grounding, personnel and worksurface controls	106–151
15:15–15:25	Afternoon break	Admin / assessment
15:25–16:45	Ionisation, shielding, packaging and material movement	106–151
16:45–17:45	Activities 3–4: EPA design and automated-cell controls	A3: 145–147, A4: 148–150
17:45–18:00	Day 1 recap and evidence check	Admin / assessment

Day 2 Schedule

Time	Session / learning activity	Slide reference
09:00–09:20	Digital attendance and Day 1 retrieval practice	Admin / assessment
09:20–10:30	Module 3: standards and programme architecture	153–178
10:30–10:40	Morning break	Admin / assessment
10:40–12:00	Measurement discipline, instruments and verification	153–178
12:00–13:00	Activities 5–6: verification round and standards audit	A5: 172–174, A6: 175–177
13:00–13:45	Lunch	Admin / assessment
13:45–14:45	Module 4: competence, incident	180–199

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	response and CAPA	
14:45–14:55	Afternoon break	Admin / assessment
14:55–15:35	Activity 7: incident investigation	A7: 193–195
15:35–16:00	Activity 8: management capstone and course summary	A8: 196–198
16:00–17:00	Written Assessment (SAQ) — 1 hour	Admin / assessment
17:00–18:00	Case Study (CS) — 1 hour	Admin / assessment

Activity and Assessment Alignment

Activity	K/A mapping	Output	Slides
1. Static Hazard Hunt at NovaSense Incoming Inspection	K1 · A1	Annotated hazard map; Risk-prioritisation worksheet; Three-control proposal; Peer-review note	99–101
2. Medical Microchip Failure Triage: ESD, EOS or Process Defect?	K1 · A1	Chronology; Hypothesis matrix; Containment plan; Decision-gate briefing	102–104
3. Design and Commission an EPA for Orion Optics	K3 · K4 · A1	EPA layout; Grounding/bonding diagram; Insulator-control register; Commissioning plan and readiness certificate	145–147
4. Select Controls for the Aster Robotics Handling Cell	K3 · K4 · A1	Exposure-path map; Control-selection matrix; Monitoring/stop-condition plan; Maintenance plan	148–150
5. Execute a Compliance Verification Round	K2 · K3 · A2	Completed measurement log; Failure/disposition records; Metric calculation; Signed evidence package	172–174
6. Standards-Based EPA Audit at Helix Assembly	K2 · A2 · A3	Audit plan; Completed checklist; Evidence register; Graded findings and follow-up plan	175–177
7. Investigate the PulseMed ESD Incident	K4 · A2 · A3	Containment record; Timeline and causal analysis; CAPA plan; Effectiveness-verification plan	193–195
8. Management Capstone: Build the Apex ESD Control Plan	K1–K4 · A1–A3	ESD control plan; Qualification and verification matrices;	196–198

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		Competency matrix; Metrics dashboard and 90-day roadmap	

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